

Assessment Tool 1

Question 1: Water Jug Problem

Problem

Two jugs of capacities **4 gallons** and **3 gallons** are given. Obtain exactly **2 gallons** in the 4-gallon jug.

Steps

Step	4-Gallon Jug	3-Gallon Jug
Initial	0	0
Fill 3-gallon jug	0	3
Pour into 4-gallon jug	3	0
Fill 3-gallon jug again	3	3
Pour into 4-gallon jug until full	4	2
Empty 4-gallon jug	0	2
Pour remaining 2 gallons into 4-gallon jug	2	0

Result

The 4-gallon jug contains **exactly 2 gallons**.

Question 2: Mars Rover Intelligent Agent

i. Percepts

- Camera images
- Rock and soil samples
- Temperature
- Terrain information
- Atmospheric data

ii. Operating Environment

- Partially observable
- Dynamic
- Sequential
- Continuous
- Unknown environment

iii. Actions

- Move forward/backward
- Turn left/right
- Capture images
- Drill rocks
- Collect samples
- Analyze samples
- Send data to Earth

iv. Performance Measures

- Successful sample collection
- Navigation accuracy
- Energy efficiency
- Mission completion
- Safe obstacle avoidance

v. Suitable Agent Architecture

Learning Agent

Reason:

- Learns from experience.
- Adapts to unknown Martian conditions.
- Improves navigation and decision-making over time.

Question 3: 8-Queens Problem

Problem Components

Initial State

- Empty 8×8 chessboard.

State Space

- All possible arrangements of 8 queens.

Operators

- Place one queen in each column.

Goal State

- No two queens attack each other.

Path Cost

- Equal cost for every move.

Search Strategy

- Backtracking Search
- Depth First Search (DFS)

Constraints

- One queen per row.
- One queen per column.
- No two queens share the same diagonal.

Question 4: OLA Cab Booking

Type of Problem-Solving Agent

Goal-Based Agent

Pseudocode

Start

Open OLA App

Enter Pickup Location

Enter Destination

Display Available Cab Types

User Selects Cab

Calculate Fare

Confirm Booking

Assign Driver

Driver Picks Up Customer

Reach Destination

End

Question 5: Uniform Cost Search (UCS)

Algorithm

1. Start from source node S.
2. Insert S into the priority queue with cost 0.
3. Remove the node with the smallest cost.
4. If destination G is reached, stop.
5. Otherwise, expand neighboring nodes.
6. Update cumulative costs.
7. Insert updated nodes into the priority queue.
8. Repeat until goal is reached.

Advantages

- Finds the least-cost path.
- Complete and optimal.
- Works well for weighted graphs.

Applications

- GPS navigation
- Logistics and delivery routing
- Network routing
- Robot path planning